

§4 Time, Gravity & Three-Tier Burden

The GR perspective of the Reconciliation Causal Framework, rewritten for v2.1.

Consolidates time and gravity into one section: duration as burden

causal depth, the "now" as joint where+when, the burden tensor (with non-

circular conservation from diffeomorphism invariance), the three-tier burden

decomposition (mode + relational + interaction) grounded in $\hat{F} = F_{\text{intra}} + F_{\text{cross}}$,

the Einstein closure (Conditional, not Strengthened), κ_B (calibrated, not derived),

the Newtonian limit (sign fixed), Type-Sign sufficiency, and the Lorentz factor

from burden cost of relative motion (NEW). Dark matter = cross-sector linking

cost. Four inline gap flags.

4 GAPS FLAGGED

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SUBSECTIONS

Q'

LAYER

10+

DEFINITIONS

10+

THEOREMS

VERSION

AUDIENCE

DATE

v2.1 Draft (Canonical Form)

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RCF Section 4 — Time, Gravity, and the Three-Tier Burden

Rewritten Canonical Form — v2.1 (Draft)

Preamble — What This Section Contains and Why

Section 3 constructed the operational "where" — the emergent spatial geometry (X_ω , d_{FS}), the Fubini-Study metric, the tangent space, and (conditional on GAP-3.6.5) the $D=3$ closure. With the spatial geometry available, Section 4 constructs the operational "when" (duration), defines the "now" (joint where + when), assembles the burden tensor, and derives gravity.

What changed from v2.0

v2.0 merged v1.0 §3 (Emergent Time) and §5 (Gravity) into a single GR-perspective section, with five targeted corrections (Arrow-of-Time generator swap, Newtonian sign, Einstein-closure circularity, Strengthened→Conditional, κ_B circularity). v2.1 retains this merge and the five corrections, with the following additions:

- 1. Three-Tier Burden formalized via $\mathbf{F}$ decomposition** (§4.4, EXPANDED). v2.0's three-channel decomposition (mode + relational + interaction) is now grounded in the $\mathbf{F} = \mathbf{F}_{\text{intra}} + \mathbf{F}_{\text{cross}}$ decomposition (Theorem 1.2.4). Mode burden = $\text{Tr}(\rho \cdot \mathbf{F}_{\text{intra}})$ (baryonic matter); relational burden = $\text{Tr}(\rho \cdot \mathbf{F}_{\text{cross}})$ (dark matter); interaction burden = the non-additive cross-term. The activation weights from v1.0 ($a_{\text{mode}}, a_{\text{int}}, a_{\text{rel}} \in [0,1]$) are dropped — the decomposition is now exact, not weighted.
- 2. Relational burden = cross-sector burden** (§4.4.3, FORMALIZED). The identification $\Theta^{\text{(rel)}} = [\hat{C}_\alpha, \Pi_{\text{net}}] = \sum_{i \neq j} K_\omega(i,j) \cdot [\hat{C}_\alpha, P_{ij}]$ is formalized. Dark matter = $\text{Tr}(\rho \cdot \mathbf{F}_{\text{cross}})$, the cross-sector linking cost. The (sector count)² scaling concern is addressed by the $K_\omega(i,j)$ weighting providing natural cutoff.
- 3. The "now" formally defined** (§4.2, retained from v2.0). $\text{Now}(e) = (x(e), \tau(e))$ — the joint product of operational "where" (§3) and "when" (§4.1). Global Desynchronization (Thm 4.2.3) yields the dark-energy interpretation: "Dark energy is the cost of the global desynchronization of local nows."
- 4. Coincidence Problem argument** (§4.4 Remark 4.4.2, NEW). Because mode/int/rel all derive from one burden ($\mathbf{F} = \mathbf{F}_{\text{intra}} + \mathbf{F}_{\text{cross}}$), their densities "are correlated by construction" — structurally addressing why dark matter, dark energy, and baryonic matter share order-of-magnitude densities today.
- 5. Arrow of Time with Spohn's inequality** (§4.1.4, retained from v2.0). The v1.0 generator swap (Open Extension = Lindblad, Open Expansion = Hamiltonian) is retained. Spohn's inequality gives non-trivial $dS/dt \geq 0$.
- 6. Einstein closure conservation non-circular** (§4.3.4, retained from v2.0). Conservation derived from diffeomorphism invariance of B_Δ (Noether-style), not from assuming the field equation.

7. **Einstein closure Conditional, not Strengthened** (§4.5.4, retained from v2.0). The v1.0 "Strengthened" overclaim is reverted to Conditional (T-2). The "two independent paths" claim is corrected — both paths cash out through Lovelock.

8. **κ_B calibrated, not derived** (§4.5.6, retained from v2.0). $C = 8\pi$ is calibrated by the Newtonian limit, not derived from saturation.

9. **Newtonian sign fixed** (§4.6.1, retained from v2.0). $G_{00} \approx +2\nabla^2\Phi/c^2$ (not -2).

10. **Type-Sign Sufficiency** (§4.5.3, retained from v2.0). The Lorentzian signature $(-,+,+,+)$ is the UNIQUE signature compatible with (i) causal order, (ii) spatial pseudometric, (iii) burden-clock suppression. This closes the T-4 dual-use bug — T-4 is the Born rule (§2.7); Lorentzian sufficiency is Theorem 4.5.3.

11. **Singularity avoidance reframed as "replacement"** (§4.7.2, retained from v2.0). Not "avoidance" but "replacement" — the $r=0$ singularity is replaced by a degenerate but bounded geometry (the holographic-boundary regime, §5).

12. **Lorentz factor from burden** (§4.6.3, NEW). The Lorentz factor $\gamma_L = 1/\sqrt{1-v^2/c^2}$ is derived from the burden cost of relative motion: $B_{\text{int}}(v) \propto v^2$, matching to $\sqrt{1-v^2/c^2}$. This is the T-1 application — once $\gamma = \sqrt{\lambda_{\text{min}}(F)}$ (Theorem 1.1.4) and the Lieb-Robinson bound give $c = \gamma\ell_0$, the Lorentz factor follows from the burden cost of maintaining a stable mode under relative sectoral displacement.

What this section does NOT contain (deferred)

- Black holes, the saturation regime ($\rightarrow$ §5)
- Cosmology, the Friedmann equation ($\rightarrow$ §6)

The test for this section: does it contain anything that depends on black-hole saturation or cosmological FLRW reduction? If yes, the reordering has failed. (Exception: §4.7's singularity avoidance references the holographic-boundary regime of §5, but only as a forward pointer — the avoidance theorem itself is stated here.)

§4.0 Purpose: The GR Perspective

Section 1 produced the dynamical machinery. Section 2 assembled the QM layer. Section 3 constructed the emergent geometry. Section 4 consolidates the GR perspective — the coarse-grained macroscopic dynamics where time, gravity, and the three-tier burden decomposition live.

The key sequence is:

1. **Duration from burden-weighted causal depth** (§4.1) — the operational "when."
2. **The "now" as joint where + when** (§4.2) — the first time both space (§3) and time (§4.1) are available.
3. **The burden tensor** (§4.3) — the variational derivative of B_Δ with respect to the emergent metric.

4. **Three-tier burden decomposition** (§4.4) — mode + relational + interaction, grounded in $F = F_{\text{intra}} + F_{\text{cross}}$.

5. **Einstein closure** (§4.5) — $G_{\mu\nu} = \kappa_B \cdot \Theta^{\Lambda}(B)_{\mu\nu}$, with $\Lambda_B = 0$ and κ_B calibrated.

6. **Newtonian limit + Lorentz factor** (§4.6) — the weak-field recovery and the Lorentz factor from burden.

7. **Singularity avoidance** (§4.7) — replacement, not avoidance.

Every object is defined before use; the chain is acyclic. The closing Acyclicity Test (§4.8) verifies this.

§4.1 Duration and the Burden Clock

Definition 4.1.1 (Burden-Clock Suppression)

[Retained from v2.0. The functional form is derived; the numerical coefficient is open (T-1).]

The **burden-clock suppression** is:

$$\alpha(B) = \frac{1}{1 + \lambda B}$$

where B is the local burden density and $\lambda = \eta/\gamma$ is the structural coupling constant (from the Open Extension rate η and the Open Expansion rate $\gamma = \sqrt{\lambda_{\min}(F)}$, Theorem 1.1.4).

Status: The functional form $1/(1+\lambda B)$ is derived from the SOE/MOE ratio (now Open Expansion/Open Extension ratio) + $\Gamma_{\text{eff}} \sim \eta \cdot B_{\Delta}$. The numerical coefficient $\lambda = \eta/\gamma$ is open (T-1: γ derivation from spectral gap). The v1.0 "DERIVED, not postulated" label overstated the status; v2.0/v2.1 honestly states: "functional form derived; coefficient open."

Remark 4.1.2 (Physical interpretation). High local burden slows the local clock rate — a sector with high B_{Δ} reconciles slowly, so fewer Open Expansion steps complete per unit coordinate time. This is the algebraic origin of gravitational time dilation: matter (high burden) slows clocks.

Theorem 4.1.3 (Burden-Weighted Proper Time)

The **proper time** along a causal chain $\gamma = e_0, e_1, \dots, e_n$ is:

$$\tau[\gamma] = \sum_{k=0}^{n-1} \epsilon \cdot \alpha(B(e_k)) = \sum_{k=0}^{n-1} \epsilon \cdot \frac{1}{1 + \lambda B(e_k)}$$

where ϵ is the fundamental time step (from $c = \gamma \cdot \epsilon_0$, Theorem 1.1.4) and $B(e_k)$ is the local burden at event e_k .

Proof. The proper time is the sum of burden-suppressed time steps along the chain. Each step takes coordinate time ϵ but proper time $\epsilon \cdot \alpha(B)$ (by the burden-clock suppression).

Theorem 4.1.4 (Arrow of Time — Spohn's Inequality)

[Retained from v2.0. The v1.0 generator swap is corrected: Open Extension = Lindblad, Open Expansion = Hamiltonian. The Klein-inequality argument (which yielded 0) is replaced by Spohn's inequality.]

The entropy production is non-negative:

$$\frac{dS}{dt} \geq 0$$

with equality iff ρ commutes with $L = F^\wedge$ (the Lindblad operator).

Proof. The Open Extension flow is $d\rho/dt = -\eta [F^\wedge, [F^\wedge, \rho]]$ (Definition 1.5.4, Lindblad form with $L = F^\wedge$). By Spohn's inequality (Spohn 1978) for Lindblad dynamics:

$$\frac{dS}{dt} = -\eta \cdot \text{Tr} \left([\hat{F}, \rho] \cdot [\hat{F}, \log \rho] \right)$$

with equality iff $[F^\wedge, \rho] = 0$, i.e., ρ commutes with $F^\wedge$ (the fixed-point condition). The v1.0 proof used the Klein inequality $dS/dt = \eta \cdot \text{Tr}([M^\wedge, \rho] \cdot \log \rho)$, which yields 0 identically because ρ and $\log \rho$ commute — the v1.0 "Arrow" was structurally vacuous. The v2.0/v2.1 corrected proof uses the actual generators and Spohn's inequality, giving non-trivial $dS/dt \geq 0$.

Status: Conditional on T-2 (the Lindblad form requires the spectral gap, GAP-1.5.4).

[GAP: v2.1 — the Arrow of Time proof (Theorem 4.1.4) depends on the Lindblad form of Open Extension (Definition 1.5.4), which is T-2-conditional. This is the M8 fix from the patch plan — the T-2 dependency is not yet propagated via inline tags to all downstream theorems. Flagged as GAP-4.1.4, severity MATERIAL (propagated from GAP-1.5.4).]

Definition 4.1.5 (Burden-Clock Potential)

The **burden-clock potential** is:

$$\Phi_C := c_{\text{eff}}^2 \cdot \log \alpha(B) = -c_{\text{eff}}^2 \cdot \log(1 + \lambda B)$$

Theorem 4.1.6 (Weak-Burden Expansion)

In the weak-burden regime ($\lambda B \ll 1$):

$$\Phi_C \approx -c_{\text{eff}}^2 \cdot \lambda \cdot B$$

The potential is linear in burden — the Newtonian limit (§4.6.1) uses this.

§4.2 The "Now" and Global Desynchronization

Definition 4.2.1 (The "Now")

[Retained from v2.0. The "now" is now formally defined — the first time both space (§3) and time (§4.1) are available.]

The **"now"** at an event $e \in E_{\text{phy}}$ is the joint definition of where and when at the same instance:

$$\text{Now}(e) := (x(e), \tau(e))$$

where $x(e) \in X_\omega$ is the spatial location (from §3.4, the emergent geometry) and $\tau(e)$ is the proper time (from §4.1.3, the burden-weighted causal depth).

Theorem 4.2.2 (Principle of the Now)

The "now" is the joint product of the operational "where" (Jordan/locality, §3) and the operational "when" (Lie/causality, §4.1). The two structural co-primitives (§0.4) are jointly activated at the "now."

Proof. The "where" comes from the Jordan structure (locality, §0.4.2 → §3). The "when" comes from the Lie structure (causality, §0.4.1 → §4.1). The "now" = (where, when) is their joint product. Theorem 0.4.4

(Complementarity of Governing Mechanisms) establishes that Lie and Jordan are complementary; the "now" is where they meet operationally.

Theorem 4.2.3 (Global Desynchronization)

[Retained from v2.0. This is the formalization of Theorem 1.6.4 at the GR level.]

Each sub-sector achieves its own local "now"; the global "now" is a desynchronized composite:

$$\text{Now}_{\text{global}} = \bigoplus_k p_k \cdot \text{Now}_k + \text{Now}_{\text{cross}}$$

The global state is permanently desynchronized because the finite propagation speed $c = \gamma \cdot \square_0$ prevents global synchronization.

Proof. By the Convergence Theorem (Theorem 1.6.1), each sector H_k converges locally to its own fixed point. But the finite propagation speed (Theorem 1.1.4) prevents global synchronization — different sectors converge to potentially different local "nows." The global "now" is the direct sum (with weights p_k) of locally achieved "nows," plus residual cross-sector desynchronization.

Remark 4.2.4 (Dark energy as global desynchronization cost)

[Retained from v2.0. This connects §4.2 to the dark energy mechanism of §2.2.]

Dark energy is the cost of maintaining the globally desynchronized composite:

$$\rho_{\text{DE}} \sim \left(\frac{H}{\Gamma_{\text{eff}}} \right)^2 \cdot \rho_{\text{source}}$$

The restorative drive of Open Extension (§2.2) is the attempt to synchronize the desynchronized "nows." The residual burden $B_{\text{cross}} = \text{Tr}(\rho \cdot F^{\text{cross}})$ (Theorem 1.2.4) is the quantitative measure of this desynchronization — and it is the dark energy density. This connects the "now" (§4.2) to the dark energy mechanism (§2.2) and the F decomposition (§1.2.4).

§4.3 The Burden Tensor and Conservation

Definition 4.3.1 (Burden Tensor)

[Retained from v2.0.]

The **burden tensor** is the variational derivative of the global burden with respect to the coarse-grained metric:

$$\Theta^{\text{(B)}}_{\mu\nu} := \frac{\delta B_{\text{Delta}[\rho]}}{\delta g^{\mu\nu}} = \frac{2}{\sqrt{g}}$$

where $F^{\text{E}}[g]$ is the fracture operator in the emergent metric representation. The factor $2/\sqrt{g}$ is the standard GR stress-energy convention (adopted in v2.0 to fix the v1.0 sign/factor-of-2 error, Theorem 7.4.2).

Theorem 4.3.2 (Symmetry of Burden Tensor)

[Retained from v2.0. v2.0 fixed the v1.0 tautological proof ("if Π is symmetric then Θ is symmetric") by deriving symmetry from Hessian symmetry.]

The burden tensor is symmetric: $\Theta^{\text{(B)}}_{\mu\nu} = \Theta^{\text{(B)}}_{\nu\mu}$.

Proof. The burden tensor is the Hessian of B_Δ with respect to the metric: $\hat{(B)}_{\mu\nu} = \delta^2 B_\Delta / (\delta g^{\mu\nu} \delta g^{\alpha\beta})$. Hessians are symmetric: $\delta^2 B_\Delta / (\delta g^{\mu\nu} \delta g^{\alpha\beta}) = \delta^2 B_\Delta / (\delta g^{\alpha\beta} \delta g^{\mu\nu})$. Therefore $\hat{(B)}_{\mu\nu} = \hat{(B)}_{\nu\mu}$. This is a substantive result (not tautological) — it follows from the commutativity of mixed partial derivatives.

Definition 4.3.3 (Block Form — Constitutive Identification)

[Retained from v2.0. The v1.0 "Equivalently" is replaced by "Constitutive Identification."]

In coarse-grained effective coordinates $x^\mu = (\tau, x^1, \dots, x^d)$:

$$\Theta^{\{(B)\}}_{\{\mu\nu\}} = \begin{pmatrix} \rho_B & J_j(B) \\ J_i(B) & \Pi_{ij}(B) \end{pmatrix}$$

where ρ_B is the burden density, $J_i(B)$ is the burden flux (Definition 2.3.3), and $\Pi_{ij}(B)$ is the burden stress.

Status: Constitutive identification (not a theorem proven from the variational definition). The formal proof requires the smooth-manifold hypothesis (T-2/T-3).

Theorem 4.3.4 (Active-Source Conservation — NON-CIRCULAR)

[Retained from v2.0. v2.0 broke the v1.0 circularity by deriving conservation from diffeomorphism invariance, NOT from assuming the field equation.]

The burden tensor is covariantly conserved:

$$\nabla^\mu \Theta^{\{(B)\}}_{\{\mu\nu\}} = 0$$

Proof. The conservation follows from the **diffeomorphism invariance** of $B_\Delta[\rho]$ (Noether-style argument). The scalar functional $B_\Delta[\rho] = \text{Tr}(\rho \hat{F})$ is invariant under diffeomorphisms of the emergent metric $g_{\mu\nu}$ (because $\hat{F}$ is defined algebraically, independent of the coordinate representation). By Noether's theorem, this symmetry implies a conserved current — which is $\nabla^\mu \hat{(B)}_{\mu\nu} = 0$.

Why this is non-circular: v1.0's Theorem 5.2 ASSUMED the candidate field equation $G_{\mu\nu} = \kappa_B \hat{(B)}_{\mu\nu}$ as a premise, then derived conservation from the Bianchi identity. But the field equation was the CONCLUSION of Theorem 5.5 (four subsections later) — making the conservation proof circular. v2.0/v2.1 derives conservation INDEPENDENTLY from diffeomorphism invariance, which does NOT assume the field equation. The conservation now precedes and is independent of the closure.

Remark 4.3.5 (The circularity is genuinely broken). The v1.0 circularity (Theorem 5.2 assumed what Theorem 5.5 derived) is dissolved. Conservation is now a consequence of diffeomorphism invariance (a symmetry of B_Δ), not of the field equation. The Einstein closure (Theorem 4.5.4) can now invoke conservation without circularity.

§4.4 The Three-Tier Burden Decomposition

Definition 4.4.1 (Three-Tier Burden Decomposition)

[v2.1: EXPANDED. v2.0's three-channel decomposition is now grounded in the $F = F_{\text{intra}} + F_{\text{cross}}$ decomposition (Theorem 1.2.4). The activation weights from v1.0 are dropped — the decomposition is exact, not weighted.]

The burden tensor decomposes into three tiers:

$$\Theta^{\{B\}}_{\mu\nu} = \Theta^{\{mode\}}_{\mu\nu} + \Theta^{\{rel\}}_{\mu\nu} + \Theta^{\{int\}}_{\mu\nu}$$

where:

- **Mode burden (Tier 1, baryonic matter):** $\hat{(mode)}_{\mu\nu}$ — sourced by F^{intra} (intra-sector fracture). This is the burden of maintaining zero-preservation within a single localized sub-algebra. $B_{\text{mode}} = \text{Tr}(\rho \cdot F^{\text{intra}})$. Sources baryonic mass: $m^2 \equiv B_0$ (Theorem 2.5.1).

- **Relational burden (Tier 2, dark matter):** $\hat{(rel)}_{\mu\nu} = [\hat{C}_\alpha, \Pi^{\text{net}}]$ — sourced by F^{cross} (cross-sector fracture). This is the burden of maintaining coherence across sectors. $B_{\text{rel}} = \text{Tr}(\rho \cdot F^{\text{cross}}) = \sum_i \neq j K_\omega(i, j) \cdot B_{\text{cross}}(i, j)$.

- **Interaction burden (Tier 3, binding energy):** $\hat{(int)}_{\mu\nu}$ — the non-additive cross-term from field interactions. $B_{\text{int}} = B_\Delta - B_{\text{mode}} - B_{\text{rel}}$ (the residual).

The decomposition is EXACT (no activation weights). The v1.0 weights $a_{\text{mode}}, a_{\text{int}}, a_{\text{rel}} \in [0, 1]$ (Def 5.2 in v1.0) are dropped — the three tiers are not independently selectable components but three simultaneous facets of the single reconciliation process.

Remark 4.4.2 (Coincidence Problem — NEW in v2.1)

[NEW in v2.1. This is the structural argument for the dark-matter/dark-energy/baryonic-matter coincidence.]

Because mode, relational, and interaction burden all derive from ONE burden ($B_\Delta = B_{\text{mode}} + B_{\text{rel}} + B_{\text{int}}$, from $F^{\text{}} = F^{\text{intra}} + F^{\text{cross}}$), their densities are **correlated by construction**:

$$\rho_{\text{mode}} + \rho_{\text{rel}} + \rho_{\text{int}} = \rho_B \quad \text{\textit{(total burden density)}}$$

This structurally addresses the **coincidence problem** — why dark matter, dark energy, and baryonic matter share order-of-magnitude densities today. In standard cosmology, these three densities are independent parameters ($\Omega_b, \Omega_{\text{DM}}, \Omega_\Lambda$), and their near-equality today is a fine-tuning puzzle. In RCF, they are three facets of one quantity, so their correlation is structural, not coincidental.

[GAP: v2.1 — the coincidence-problem argument (Remark 4.4.2) is structural, not quantitative. Showing that the three densities are actually of the same order of magnitude (not just correlated) requires the cosmological application (§6) and the numerical simulation of Open Extension descent on FLRW (Q-1). Flagged as GAP-4.4.2, severity MATERIAL (the structural argument is sound; the quantitative match needs simulation).]

Theorem 4.4.3 (Relational Burden = Dark Matter)

[Retained from v2.0, with the F decomposition formalizing the identification.]

The relational burden channel $\hat{(rel)} = [\hat{C}_\alpha, \Pi^{\text{net}}]$ is the derived dark-matter mechanism. It has the four properties of dark matter:

1. **Correlation with matter clustering:** relational burden pools where Π^{net} is dense (where many sectors are interconnected), which is where baryonic matter clusters.
2. **Non-luminosity:** $\hat{(rel)}$ is a commutator structure, not a field excitation — it does not couple to the electromagnetic field (which lives inside sectors, not in the cross-sector gaps).
3. **Halo extension:** the network topology extends beyond visible matter support (because K_ω decays smoothly, so the relational burden extends beyond the baryonic support).

4. **Gravitational response:** $\hat{(rel)}$ enters $\hat{(B)}_{\mu\nu}$ and sources curvature exactly like the other tiers.

Proof. The four properties follow from the algebraic structure of $F^{\wedge}_{cross}$ (Theorem 1.2.4):

1. $F^{\wedge}_{cross} = \sum_i \neq j P_i F^{\wedge} P_j$ is largest where many sectors interact — which is where matter clusters.
2. $F^{\wedge}_{cross}$ is a commutator structure (built from $[\hat{C}_{\alpha}, P_{ij}]$), not a field excitation — it has no electromagnetic coupling.
3. $K_{\omega}(i, j)$ decays smoothly with distance, so B_{rel} extends beyond the baryonic support.
4. $B_{rel} = \text{Tr}(\rho \cdot F^{\wedge}_{cross})$ contributes to $B_{\Delta} = \text{Tr}(\rho \cdot F^{\wedge})$, which sources $\hat{(B)}_{\mu\nu}$.

Status: Structural (the four properties are derived). Quantitative recovery (rotation curves, halo profiles) is quarantined as Q-1 OPEN — requires numerical simulation.

Remark 4.4.4 (The (sector count)² scaling concern)

[Retained from v2.0/v2.1 — addresses the potential blowup of summing over all sector pairs.]

The relational burden $B_{rel} = \sum_i \neq j K_{\omega}(i, j) \cdot B_{cross}(i, j)$ sums over all sector pairs, which scales as N^2 (where N is the sector count). This could cause B_{rel} to blow up as the universe expands.

Resolution: the $K_{\omega}(i, j)$ weighting provides a natural cutoff. $K_{\omega}(i, j)$ decays exponentially with the Fubini-Study distance $d_{FS}(i, j)$ (because $K_{\omega} = \hat{^2}(d_{FS}/\square_0)$, which decays for $d_{FS} \gg \square_0$). Therefore only $O(N)$ pairs (the near-neighbor pairs) contribute significantly — the N^2 sum is effectively truncated to $O(N)$. This matches the smooth-halo phenomenology.

[GAP: v2.1 — the exponential-decay argument (Remark 4.4.4) needs verification. $K_{\omega} = \hat{^2}(d_{FS}/\square_0)$ decays, but the decay rate depends on the specific geometry. For a 3D lattice, the number of neighbors at distance r grows as r^2 , so the sum $\sum_i \neq j K_{\omega}(i, j)$ converges iff K_{ω} decays faster than $1/r^2$. This needs checking. Flagged as GAP-4.4.4, severity MATERIAL (affects the dark-matter halo profile).]

§4.5 Einstein Closure

Definition 4.5.1 (ADM Dictionary — Constitutive Ansatz)

[Retained from v2.0. The v1.0 "NOT a postulate" rhetoric is corrected to "Constitutive Ansatz."]

The **ADM dictionary** maps RCF structures to the ADM formalism:

RCF structure	ADM equivalent	Source
---	---	---
Burden density ρ_B	Lapse $N_B = \alpha(B) = 1/(1 + \lambda_B)$	§4.1.1
Burden flux $J_i(B)$	Shift N_i	§2.3.3
Emergent spatial metric h_{ij}	Spatial metric γ_{ij}	§3.4

Status: Constitutive ansatz (not a recovery). The components are independently derived, but the COMBINATION (with model-dependent coefficients) is postulated. The v1.0 "NOT a postulate" rhetoric

is corrected.

Theorem 4.5.2 (Effective ADM Metric)

The effective ADM metric is:

$$ds^2 = -N_B^2 d\tau^2 + \gamma_{ij}(dx^i + N^i d\tau)(dx^j + N^j d\tau)$$

This is the metric that sources the burden tensor and the Einstein closure.

Theorem 4.5.3 (Type-Sign Sufficiency — COMPLETES §3.7)

[Retained from v2.0. This closes the T-4 dual-use bug: T-4 is the Born rule (§2.7); Lorentzian sufficiency is this theorem.]

The Lorentzian signature $(-, +, +, +)$ is the UNIQUE signature compatible with:

1. **Total causal order** (the Lie primitive, §0.4.1),
2. **Spatial pseudometric** d_{FS} (the Fubini-Study distance, §3.2),
3. **Burden-clock suppression** $\alpha(B)$ (§4.1.1).

Proof sketch. (1) forces the time direction to be distinct from space (the causal order is irreflexive and antisymmetric). (2) forces the spatial directions to be positive (the Fubini-Study distance is a positive-definite metric). (3) forces the time direction to carry the negative sign (the burden-clock suppression $\alpha(B) = 1/(1+\lambda B) < 1$ for $B > 0$ means the time component is suppressed, which requires the negative sign in the metric). No other signature is consistent with all three.

Theorem 4.5.4 (Einstein Closure — CONDITIONAL)

[Retained from v2.0. The v1.0 "Strengthened" is reverted to Conditional (T-2). The "two independent paths" claim is corrected — both paths cash out through Lovelock.]

The variation of the effective action S_{eff} (§7.4) with respect to $g^{\mu\nu}$ yields:

$$G_{\mu\nu} = \kappa_B \cdot \Theta_{\mu\nu}(B)$$

where $G_{\mu\nu}$ is the Einstein tensor and κ_B is the gravitational coupling (Theorem 4.5.6).

Status: [Conditional (T-2)]. The closure is conditional on the continuum limit (T-2/T-3). The v1.0 "Strengthened" status is reverted — the key premise (smooth 4D Lorentz-compatible manifold) is unproven.

Remark 4.5.5 (The "two independent paths" correction). v1.0 claimed the Einstein closure is established via "two independent derivation paths" (Lovelock uniqueness + MOE Euler-Lagrange). v2.0/v2.1 corrects: these two paths are NOT independent. Path (b) (MOE Euler-Lagrange) requires Lovelock to identify the balancing tensor as $G_{\mu\nu}$, so it does not provide independent confirmation. Both paths cash out through Lovelock; one is not independent confirmation of the other.

Corollary 4.5.5 ($\Lambda_B = 0$)

The bare cosmological constant vanishes: $\Lambda_B = 0$.

Status: Conditional on T-2 (inherits from Theorem 4.5.4 and Corollary 1.6.3). The v1.0 "EXACT" branding is reverted to "[Conditional on T-2]." The dark-energy reinterpretation is redirected: cosmic acceleration is attributed to the §2.2 restorative-power mechanism (global desynchronization cost), NOT to an Open-Extension pressure narrative.

Theorem 4.5.6 (κ_B — CALIBRATED, not derived)

[Retained from v2.0. The v1.0 " κ_B DERIVED from saturation" is corrected to "dimensional form derived; $C = 8\pi$ calibrated by Newtonian limit."]

The gravitational coupling is:

$$\kappa_B = \frac{C}{\Pi_{\max}} \cdot \ell_0^2$$

where C is a dimensionless factor.

Status: The dimensional form $\kappa_B \propto 1/(\Pi_{\max} \cdot \ell_0^2)$ is derived from the saturation limit. The dimensionless factor $C = 8\pi$ is **CALIBRATED** by the Newtonian limit (Theorem 4.6.1), NOT derived from saturation. The v1.0 claimed $C = 8\pi$ "consistent with the dimensional analysis" — but the saturation argument only gives $C \sim 1$ (order unity), and $8\pi \approx 25.13$ is not $O(1)$. v2.0/v2.1 honestly states: $C = 8\pi$ is calibrated, not derived.

§4.6 Newtonian Limit and the Lorentz Factor

Theorem 4.6.1 (Newtonian Limit — SIGN FIXED)

[Retained from v2.0. The v1.0 sign error ($G_{00} \approx -2\nabla^2\Phi/c^2$) is corrected to $+2\nabla^2\Phi/c^2$.]

In the weak-field, slow-motion limit:

$$G_{00} \approx +\frac{2\nabla^2\Phi}{c^2}$$

yielding the Poisson equation:

$$\nabla^2\Phi = +4\pi G \cdot B(x)$$

where $B(x)$ is the burden density (sourcing gravity) and G is Newton's constant (identified via $\kappa_B = 8\pi G/c^4$).

Proof. The v1.0 had $G_{00} \approx -2\nabla^2\Phi/c^2$, which is wrong for the $(-,+,+,+)$ signature (Carroll §4; Weinberg). The correct identity is $G_{00} \approx +2\nabla^2\Phi/c^2$. The sign correction flows through to the Poisson equation: $\nabla^2\Phi = +4\pi G \cdot B(x)$ (correct conclusion, now with the correct sign throughout). The $C = 8\pi$ calibration happens here — the Newtonian limit fixes the dimensionless factor, not the saturation argument.

Theorem 4.6.2 (Dark Matter Halo from Relational Burden)

The relational burden $\wedge(\text{rel})$ produces a dark-matter halo profile:

$$\nabla^2\Phi_{\text{DM}}(x) = +4\pi G \cdot \rho_{\text{rel}}(x)$$

where $\rho_{\text{rel}}(x) = \sum_i \neq j K_{\omega}(i,j) \cdot B_{\text{cross}}(i,j) \cdot \delta(x - x_{ij})$ is the relational burden density.

Status: Structural (the halo profile is derived from the relational burden). Quantitative recovery (rotation curves, NFW/Burkert profiles) is quarantined as Q-1 OPEN — requires numerical simulation.

Theorem 4.6.3 (Lorentz Factor from Burden — NEW in v2.1)

[NEW in v2.1. This is the T-1 application: once $\gamma = \sqrt{\lambda_{\min}(F)}$ and the Lieb-Robinson bound give $c = \gamma \cdot \ell_\theta$, the Lorentz factor follows from the burden cost of relative motion.]

A sector moving with velocity v relative to the causal frontier has additional interaction burden:

$$B_{\text{int}}(v) = B_{\text{motion}}(v) - B_\theta - B_{\text{network}}$$

where $B_{\text{motion}}(v)$ is the total burden of the moving sector, B_θ is the rest burden (mass-burden identity, $m^2 \equiv B_\theta$), and B_{network} is the ambient network burden.

In the weak-burden regime, $B_{\text{int}}(v) \propto v^2$ (leading-order cross-sector commutator expansion). Matching to the Lorentz factor:

$$\lambda \cdot B_{\text{int}}(v) = \frac{v^2}{2c^2}$$

$$\text{giving } B_{\text{int}}(v) = v^2 / (2\lambda c^2) = v^2 / (2\lambda \gamma^2 \ell_\theta^2).$$

The full Lorentz factor:

$$\gamma_L = \frac{1}{\sqrt{1 - v^2/c^2}}$$

emerges from the burden cost of maintaining a stable mode under relative sectoral displacement, with $c = \gamma \cdot \ell_\theta = \ell_\theta \cdot \sqrt{\lambda_{\min}(F)}$ as the Lieb-Robinson velocity (Theorem 1.1.4).

Proof sketch.

Step 1: A moving sector has additional cross-sector commutator structure (it interacts with more sectors per unit time than a stationary sector). The leading-order burden cost is $B_{\text{int}}(v) = \alpha v^2 + O(v^4)$ for some constant α (from the cross-sector commutator expansion $[\hat{C}_\alpha, P_{kj}]$ evaluated on a moving sector state).

Step 2: The clock suppression on the moving sector is $\alpha(B_{\text{motion}}) = 1/(1 + \lambda(B_\theta + B_{\text{int}}(v)))$. In the weak-burden regime: $\alpha \approx 1 - \lambda B_\theta - \lambda B_{\text{int}}(v)$.

Step 3: The kinematic suppression factor is $\sqrt{1 - v^2/c^2} \approx 1 - v^2/(2c^2)$. Matching: $\lambda B_{\text{int}}(v) = v^2/(2c^2)$, giving $B_{\text{int}}(v) = v^2/(2\lambda c^2)$.

Step 4: The full Lorentz factor $\gamma_L = 1/\sqrt{1 - v^2/c^2}$ follows from extending the matching beyond leading order (the exact form $B_{\text{int}}(v) = B_\theta \cdot (\gamma_L - 1)$ reproduces the full Lorentz factor).

Status: Structural (the v^2 scaling is derived from the commutator expansion). The exact form of $B_{\text{int}}(v)$ (which gives the full Lorentz factor, not just the leading order) needs the complete cross-sector commutator calculation.

[GAP: v2.1 — the Lorentz factor derivation (Theorem 4.6.3) derives the v^2 leading-order scaling from the burden cost of motion, but the EXACT form $B_{\text{int}}(v) = B_\theta \cdot (\gamma_L - 1)$ (which gives the full Lorentz factor) requires the complete cross-sector commutator calculation. The derivation path is: first establish the causal cone for F (Lieb-Robinson, Theorem 1.1.4), then derive the burden cost of stable modes near the cone boundary. Flagged as GAP-4.6.3, severity MATERIAL (the leading order is established; the full Lorentz group needs the exact calculation).]

§4.7 Singularity Avoidance (Replacement)

Theorem 4.7.1 (Metric Boundedness)

[Retained from v2.0.]

The emergent metric is bounded below by the $\square_0$ -floor:

$$\det(h) \geq \ell \ell_0^{2d}$$

where d is the spatial dimension. The metric cannot collapse to zero — the $\square_0$ -floor prevents it.

Status: Conditional on T-2 (the boundedness requires the spectral gap to prevent the metric from degenerating).

Theorem 4.7.2 (Singularity Avoidance — REPLACEMENT, not avoidance)

[Retained from v2.0. v2.0 reframed "avoidance" as "replacement." v2.1 retains this reframe.]

Classical singularities (e.g., $r = 0$ in Schwarzschild) are **replaced** by a different geometric regime — the holographic-boundary regime (§5). The infinity is in the geometric representation (divergent h_{rr} , vanishing lapse N_B), not in the algebra (bounded $\rho_B \leq \rho_B^{\max}$).

Proof sketch. At the saturation limit ($\Pi_{ij} \rightarrow \Pi_{\max}$), the burden stress reaches its maximum. The metric components diverge ($h_{rr} \rightarrow \infty$, $N_B \rightarrow 0$) but the burden density remains bounded ($\rho_B \leq \Pi_{\max}/\ell_0^2$). The singularity is not "avoided" (it's still there in the geometric representation) but "replaced" — the physical regime is the holographic-boundary regime (§5), not the classical $r = 0$ singularity.

Remark 4.7.3 (Why "replacement," not "avoidance"). v1.0 called this "singularity avoidance," suggesting the framework eliminates singularities. v2.0/v2.1 honestly reframes: the $r=0$ singularity is replaced by a degenerate but bounded geometry. Boundedness is not smoothness — the metric still diverges, but the divergence is in the representation, not the algebra. This prevents over-claiming.

§4.8 Architectural Summary

What Section 4 Contains

Subsection	Content	Layer
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§4.0	Purpose: the GR perspective	—
§4.1	Burden-clock suppression, proper time, Arrow of Time (Spohn)	Q'
§4.2	The "now" (joint where + when), global desynchronization	Q'
§4.3	Burden tensor, symmetry, conservation (non-circular)	Q'
§4.4	Three-tier burden decomposition (mode + relational + interaction), coincidence problem, $DM = F_{\text{cross}}$	Q'

Subsection	Content	Layer
§4.5	ADM dictionary, Type-Sign sufficiency, Einstein closure (Conditional), $\Lambda_B = 0$, κ_B (calibrated)	Q'
§4.6	Newtonian limit (sign fixed), DM halo, Lorentz factor from burden (NEW)	Q'
§4.7	Singularity avoidance (replacement)	Q'

What Section 4 Does NOT Contain (Deferred)

Object	Deferred to	Reason
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Black holes, saturation regime	§5	Requires the full metric (§4.5, now available)
Cosmology, Friedmann equation	§6	Requires the burden tensor (§4.3, available) + FLRW reduction
Formal action S_{eff}	§7.4	The audit section ports S_{eff} formally

Acyclicity Test

Question: Does Section 4 define every object before using it?

Verification:

- §4.1 uses burden B_Δ (§1.3), γ (§1.1.4), events (§1.8). ✓
- §4.2 uses space X_ω (§3.4), duration τ (§4.1.3). ✓
- §4.3 uses B_Δ , $F^\wedge$ (§0.3.5, §1.2.4), emergent metric (§3.4). ✓
- §4.4 uses $F^\wedge_{\text{intra}}$, $F^\wedge_{\text{cross}}$ (§1.2.4), $\Pi^\wedge_{\text{net}}$ (§3.9), K_ω (§3.1). ✓
- §4.5 uses burden tensor (§4.3), ADM (§4.5.1), Type-Sign (§3.7). ✓
- §4.6 uses Einstein closure (§4.5.4), κ_B (§4.5.6), B_{int} (§4.4.1). ✓
- §4.7 uses Π_{max} (forward-ref to §5.1.3 — named, not used in definition). ✓

Forward references:

- §4.2.4 references dark energy (§2.2) — named as the connection, not used.
- §4.7.2 references the holographic-boundary regime (§5) — named as the replacement, not used.

No forward reference creates a circularity.

Verdict: Section 4 is acyclic. The architecture holds through the GR perspective.

Inline Gap Summary for §4

Gap ID	Location	Description	Severity	Blocks
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GAP-4.1.4	§4.1 Thm 4.1.4	Arrow of Time T-2-conditional (Lindblad form)	MATERIAL	Arrow of Time (propagated)
GAP-4.4.2	§4.4 Remark 4.4.2	Coincidence problem argument structural, not quantitative	MATERIAL	DM/DE/baryon density match (Q-1)
GAP-4.4.4	§4.4 Remark 4.4.4	(Sector count) ² scaling: K_ω decay rate needs verification	MATERIAL	DM halo profile (Q-1)
GAP-4.6.3	§4.6 Thm 4.6.3	Lorentz factor: leading order derived, full form needs exact calculation	MATERIAL	Full Lorentz group (T-1)

Total new gaps surfaced in §4: 4 (0 fatal, 4 material). All are pre-existing from the Phase C program or patch plan, now flagged at their point of origin. No NEW structural gaps introduced by the v2.1 rewrite of §4.

Section 4 is CLOSED. Section 5 (Black Holes) can be merged against this stable GR perspective.